

Class Programming Homework (search algorithm) (worth 10% final grade)

Due March 1, 2026

One student could do the project along.

Two student could team up to do the project.

Students are required to write python codes of

BFS (if his/her ID is even, and this is one person team)

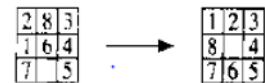
DFS (if his/her ID is odd, and this is one-person team)

If the team has two students, need to do both BFS and DFS.

All instructions and codes are from the website:

<https://www.geeksforgeeks.org/artificial-intelligence/8-puzzle-problem-in-ai/>

Student also need to get the A* code from this site.



Main task: find a path from start state to goal state on the 8-puzzle problem:

Representation of the 8-puzzle. Use 0 to denote the empty spot. Interpret 01234567 as

0 1 2

3 4 5

6 7 8.

Thus the start and goal states of the above 8-puzzle problem are 283164705 and 123804765

We need to find the path:

2 8 3

1 6 4

7 0 5

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1 2 3

8 0 4

7 6 5

Tasks: A1, B1, A2, B2.

A1. Run BSF (or DFS) (1-person team), Print the number of moves and the path.

Run both (2-person team), find the path from start state to goal state. Print # of moves and path.

If you are 2-person team, you need to do both BSF and DFS.

B1. Run A* on the same 8-puzzle problem as in (A). Print # of moves and the path

C1. Test **parity** on 8-puzzle problem. Exchange first two numbers (2 and 8) in the start state, run BFS or DFS, you will see that you cannot reach the goal state, because the two states now have different parity.

Tasks A2 and B2 are the same as A1 and B1, except the start and goal states are two random permutations of 012345678 (Use student's ID to initialize random number generator, so different team run on different start and goal states). If the start state cannot reach the goal state, you exchange the first two or any neighboring 2 numbers (see **C1**).

Print the start/goal states, # of moves and the path you find in a report. Write your name and ID on the report (if 2-person team, only One report, put both your names and ID in the report)

Each team tar all codes and a written report. Submit on blackboard.